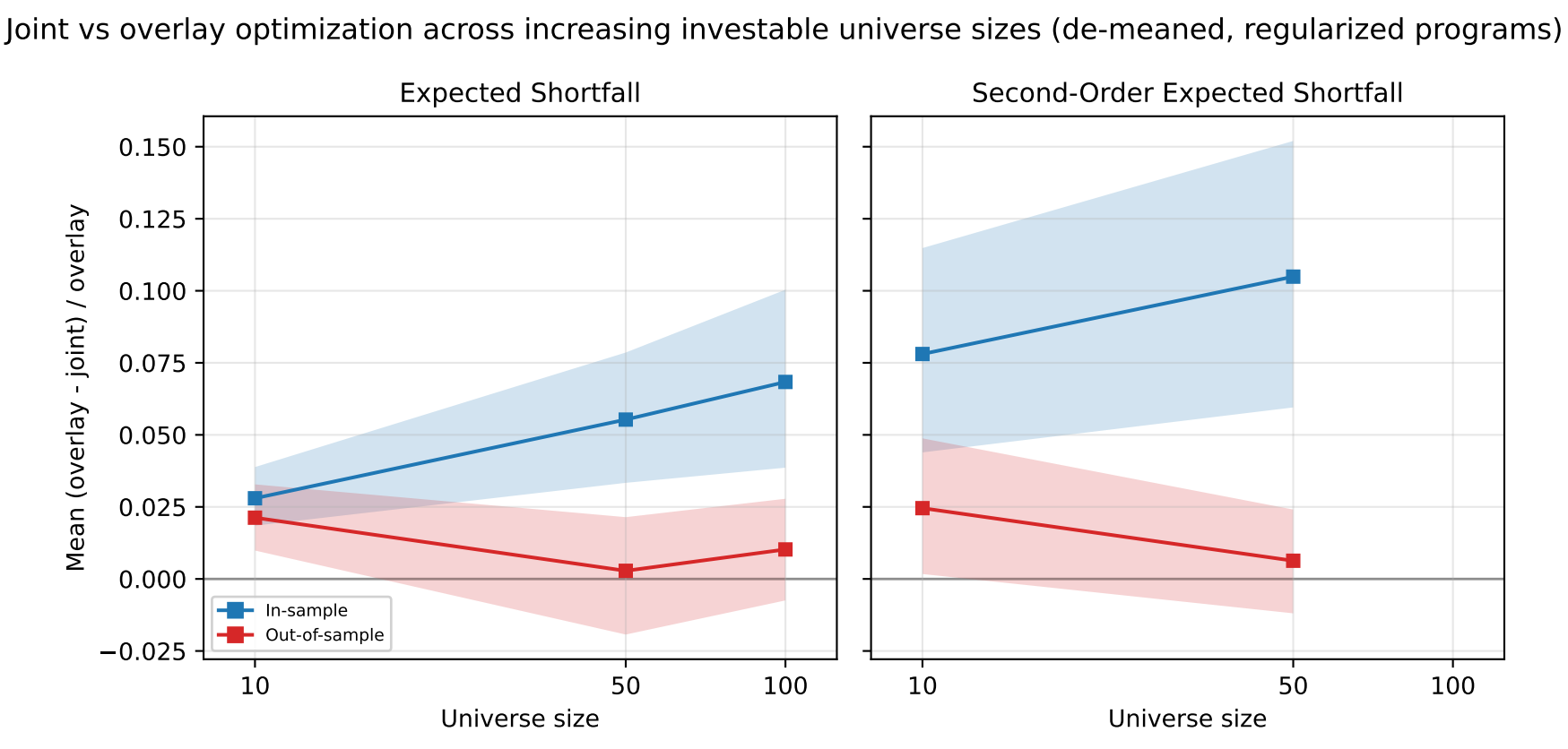
Joint vs overlay optimization across increasing investable universe sizes (de-meaned, regularized programs)
Expected Shortfall
Second-Order Expected Shortfall
Mean (overlay - joint) / overlay
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−0.025
In-sample
Out-of-sample
10
50
100
Universe size
10
50
100
Universe size